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A Genetic Modifier of the Gut Microbiome Influences the Risk of Graft-versus-Host Disease and Bacteremia After Hematopoietic Stem Cell Transplantation



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ABSTRACT

The human gut microbiome is involved in vital biological functions, such as maintenance of immune homeostasis and modulation of intestinal development and enhanced metabolic capabilities. Disturbances of the intestinal microbiota have been associated with development and progression of inflammatory conditions, including graft-versus-host disease (GVHD). The *fucosyltransferase 2* (*FUT2*) gene produces an enzyme that is responsible for the synthesis of the H antigen in body fluids and on the intestinal mucosa. *FUT2* genotype has been shown to modify the gut microbiome. We hypothesized that *FUT2* genotype influences risk of GVHD and bacterial translocation after allogeneic hematopoietic stem cell transplantation (HSCT). *FUT2* genotype was determined in 150 consecutive patients receiving allogeneic HSCT at our center. We abstracted clinical characteristics and outcomes from the transplantation database. Cumulative risk of any acute GVHD varied by *FUT2* genotype, with decreased risk in those with A/A genotype and increased risk in those with G/G genotype. In contrast, the cumulative incidence of bacteremia was increased in those with A/A genotype. We conclude that the *FUT2* genotype influences risk of acute GVHD and bacteremia after HSCT. We hypothesize that the mechanisms involve altered intestinal surface glycosylation and microbial composition but this requires additional study.

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INTRODUCTION

Recent studies have shown that intestinal inflammation secondary to graft-versus-host disease (GVHD) is associated with major shifts in the composition of the intestinal microbiota [1]. We hypothesized that *fucosyltransferase 2* (*FUT2*) genotype, a genetic modifier of the gut microbiome, influences the risk of GVHD and bacterial translocation after allogeneic hematopoietic stem cell transplantation (HSCT).

The mucin layer overlying the intestinal epithelium has an important role as the first line of host defense and enables contact between the intestinal microbiota and the host [2–4]. In a healthy microbial community, there is an optimal balance of organisms that provide signals to the developing immune system, leading to a balance of Treg and TH17 cell

activities [5]. The *FUT2* gene regulates expression of ABH blood group antigens in the mucin layer [6]. ABH mucosal antigens are carbohydrates that serve as a bacterial energy source [7] and as adhesion receptors for many microbes [8], and they play an important role in shaping the microbiota of the host [9–11]. A common nonsense mutation (428G > A) in the *FUT2* gene means that about 25% of Caucasians who have an AA genotype do not express ABH antigens in the mucin layer; these people are considered *nonsecretors* [12,13]. The potential importance of *FUT2* genotype and disease is illustrated by large GWAS (genome-wide association study) that reported a significant association between risk of Crohn's disease, differences in the gut microbiome, and *FUT2* nonsecretor status [14].

Our first hypothesis was that *FUT2* genotype would modify risk of GVHD. GVHD is initiated by translocation of lipopolysaccharide (LPS) across the gastrointestinal lining disrupted by chemotherapy, radiation, and/or infection [15]. Systemic LPS triggers secretion of proinflammatory cytokines, causing T cell attack and tissue damage typical of

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GVHD of the skin, gut, and liver. Recent murine data have demonstrated differential response to systemic LPS in secretor and nonsecretor mice, with reduced ability to recover from a systemic LPS challenge in the nonsecretor animals [16], leading us to expect that risk of GVHD would be increased in nonsecretor transplant recipients.

Our second hypothesis was that FUT2 genotype would modify risk of bacteremia. Many bacteria and fungi initiate infection by noncovalent binding to mucosal cell surface carbohydrate-binding proteins expressed by FUT2. Nonsecretor status has been associated with increased risk of a number of infections, including recurrent bacterial (pneumococcus, meningococcus, and haemophilus influenza) and fungal infections, including oral infections, urinary tract infections, and gastroduodenal disease [17–21]. In addition, in murine studies, LPS challenge led to significantly increased expression of microbial virulence genes in FUT2-negative but not FUT2-sufficient mice [16]. These data led us to hypothesize that the frequency of invasive bacterial infections would be increased in nonsecretor transplant recipients.

Our data confirm an increased risk of bacteremia in nonsecretor recipients, but in contrast to our expectations, risk of GVHD was reduced in nonsecretors.

METHODS

Patients and Transplantation Procedure

One hundred fifty consecutive patients receiving allogeneic HSCT at our center between 2010 and 2014 were included in this study. The study was approved by the institutional review board and all participants gave written informed consent. We abstracted clinical characteristics and outcomes from the transplantation database.

Study Population

Patient characteristics are summarized in Table 1. All patients received piperacillin/tazobactam as empiric antibiotic therapy with episodes of fever, and vancomycin was added with deterioration of clinical status.

Donor Selection

Donor characteristics are summarized in Table 1. High-resolution typing was performed for all donor-recipient pairs [22]. Among donors, 29% were matched sibling donors (n = 43), 49% matched unrelated or other family member donors (n = 74) and 22% were mismatched donors (n = 33).

Genotyping

Peripheral blood neutrophils were obtained before transplantation. Using the single nucleotide polymorphism rs601338 (428G > A), secretor status was genotyped. DNA was extracted from patient neutrophils with a Qiagen Gentra Puregene Blood Kit (Qiagen, Hilden, Germany), and the quantity and quality were analyzed on a NanoDrop 2000 Spectrophotometer. The DNA was amplified by PCR in a total reaction volume of 20 μ L containing Taqman 2x PCR Master Mix (4304437; Applied Biosystems,

Table 1
Characteristics of Patients who Underwent Allogeneic HSCT

Characteristic	A/A	A/G	G/G	P Value
No. of patients	34 (23%)	78 (52%)	38 (25%)	
Age at transplantation, median (range), yr	5.27 (.3–29.78)	6.55 (.41–28.67)	6.61 (.72–32.79)	.54
Male sex	24 (70%)	49 (63%)	24 (63%)	.77
Race				N/A*
Caucasian	32 (94%)	69 (88%)	29 (76%)	
African American	2 (6%)	7 (9%)	6 (16%)	
Mixed	0	2 (3%)	1 (2.6%)	
Asian	0	0	1 (2.6%)	
Pacific Islander	0	0	1 (2.6%)	
Ethnicity				.38
Non-Hispanic	33 (97%)	75 (96%)	34 (89%)	
Hispanic	1 (3%)	3 (4%)	4 (11%)	
Diagnosis				N/A*
Hematologic malignancies	10 (29%)	20 (26%)	15 (40%)	
Immune deficiencies	11 (32%)	24 (30%)	8 (21%)	
BM failure syndromes	9 (26%)	17 (22%)	6 (16%)	
Hemophagocytic lymphohistiocytosis	2 (6%)	12 (15%)	4 (10%)	
Metabolic diseases	1 (3%)	2 (3%)	4 (10%)	
Hemoglobinopathies	0	3 (4%)	1 (3%)	
Evans syndrome	1 (3%)	0	0	
Conditioning regimen				.05
Myeloablative	16 (47%)	38 (49%)	27 (71%)	
Reduced intensity	18 (53%)	40 (51%)	11 (29%)	
Graft source				.36
BM	30 (88%)	63 (81%)	29 (76%)	
PBSC	3 (9%)	10 (13%)	3 (8%)	
Cord blood	1 (3%)	4 (5%)	6 (16%)	
Cord blood and BM	0	1 (1%)	0	
HLA match				.65
Matched sibling donor	11 (32%)	19 (24%)	13 (34%)	
Matched family or unrelated donor	15 (44%)	43 (55%)	16 (42%)	
Mismatched donor	8 (24%)	16 (21%)	9 (24%)	
Use of T cell antibodies	16 (47%)	36 (46%)	10 (26%)	.09
Sex mismatch	14 (41%)	36 (46%)	21 (55%)	.46
CMV status (donor/recipient)				.44
Neg/neg	3 (9%)	13 (17%)	9 (24%)	
Neg/pos	14 (41%)	23 (29%)	15 (39%)	
Pos/neg	2 (6%)	9 (17%)	2 (5%)	
Pos/pos	15 (44%)	33 (42%)	12 (32%)	
Malignant disease status				.35
CR1 or CR2	4 (40%)	12 (60%)	9 (60%)	
Other	6 (60%)	8 (40%)	6 (40%)	

Data presented are n (%), unless otherwise indicated.

NA indicates not available; BM, bone marrow; PBSC, peripheral blood stem cells; CMV, cytomegalovirus; neg, negative; pos, positive; CR, complete remission.

* Not applicable, small sample size per cell precludes computing of meaningful P value.

Foster City, CA) and 20x *FUT2* primer/probe mix (Applied Biosystems; rs601338). The reaction was performed in an Applied Biosystems GeneAmp PCR System 9700 thermocycler under the following conditions: initial denaturation for 2 minutes at 50°C and 10 minutes at 95°C, followed by 15 seconds at 95°C and 1 minute at 60°C for 40 cycles and a hold cycle at 22°C. After PCR was complete, allelic discrimination was analyzed for each sample on an Applied Biosystems 7300 RT PCR System with 7300 system software.

The individuals with the homozygote A/A genotype were defined as *nonsecretors* and those with A/G and G/G genotypes were defined as *secretors*. Genetic analysis revealed 23% recipients had A/A genotype ($n = 34$), 52% had A/G genotype ($n = 78$), and 25% had the G/G genotype ($n = 38$); Hardy Weinberg equilibrium was confirmed.

Clinical Endpoints

GVHD

Acute GVHD was graded according to the Glucksberg system [23]. The maximum grade through day 100 was considered in this study. Biopsies were obtained when clinically indicated.

Bacteremia

Bacteremia was defined as any positive blood culture and we did not consider any positive blood culture as contaminant. We excluded all coagulase-negative isolates from the analysis and only included translocated bacteremia episodes of oral and colonic sources, including gram-negative and gram-positive organisms. A small number of patients had multiple episodes of bacteremia; however, only the first episode was considered in this study.

Statistical Analysis

Median and interquartile range and frequencies were used to describe continuous and categorical variables, respectively. Univariate and multivariate competing risk analysis were performed using the Fine and Gray method [24]. For each patient, the earlier of acute GVHD, second transplantation, death, and relapse were considered the event for the competing risk analysis. The log-rank test was used to assess the differences between genotypes with respect to overall survival. The Kaplan-Meier method was used to produce the overall survival and competing risk curves. Analyses were performed using R version 3.0.1. Statistical tests were 2-sided and significance was assessed at $P < .05$.

RESULTS

Acute GVHD

Acute GVHD occurred in 36% of patients ($n = 54$), with median day of onset of 39 days. Fifty-five percent of patients

had skin GVHD only ($n = 30$); 13% had skin, gut, and liver GVHD ($n = 7$); 11% had gut and skin GVHD ($n = 6$); 11% had gut GVHD only ($n = 6$); 6% had gut and liver GVHD ($n = 3$); and 4% had skin and liver GVHD ($n = 2$). Skin GVHD grade was 4 in 13% of patients ($n = 6$), 3 in 16% ($n = 7$), 2 in 31% ($n = 14$), and 1 in 40% ($n = 18$). Gut GVHD grade was 4 in 38% of patients ($n = 8$), 3 in 29% ($n = 6$), 2 in 29% ($n = 6$), and 1 in 4% ($n = 1$). Liver GVHD grade was 4 in 25% of patients ($n = 3$), 3 in 41% ($n = 5$), 2 in 17% ($n = 2$), and 1 in 17% ($n = 2$). Cumulative risk of any acute GVHD varied by *FUT2* genotype, with decreased risk in those with A/A genotype and increased risk in those with G/G genotype ($P = .04$) (Figure 1A). Median day of onset of acute GVHD did not differ by genotype (45.5, 39, and 39.5 days for A/A, A/G, and G/G, respectively). A/A genotype (odds ratio [OR], .4; $P = .046$), myeloablation (OR, .99; $P = .029$), and matched sibling donor (OR, .41; $P = .026$) were identified to be significant GVHD risk factors in multivariate analysis (Table 2).

Bacteremia

Bacteremia occurred in 26% of patients ($n = 40$). Fifty-three percent (53%) of patients had gram-negative bacteremia ($n = 21$) and 47% had gram-positive bacteremia ($n = 19$). In contrast to our findings in GVHD, cumulative incidence was increased in those with A/A genotype ($P = .048$) (Figure 1B). A/A genotype (OR, 3.94; $P = .0047$) and A/G genotype (OR, 2.46; $P = .05$) were associated with increased risk of bacteremia in multivariate analysis (Table 3).

Norovirus and Clostridium Difficile Infections

Norovirus infection was identified in 11% of patients ($n = 17$), while *C. difficile* infection occurred in 17% of patients ($n = 25$). There was no significant difference in incidence of norovirus or *C. difficile* infections among those with different *FUT2* genotypes (data not shown).

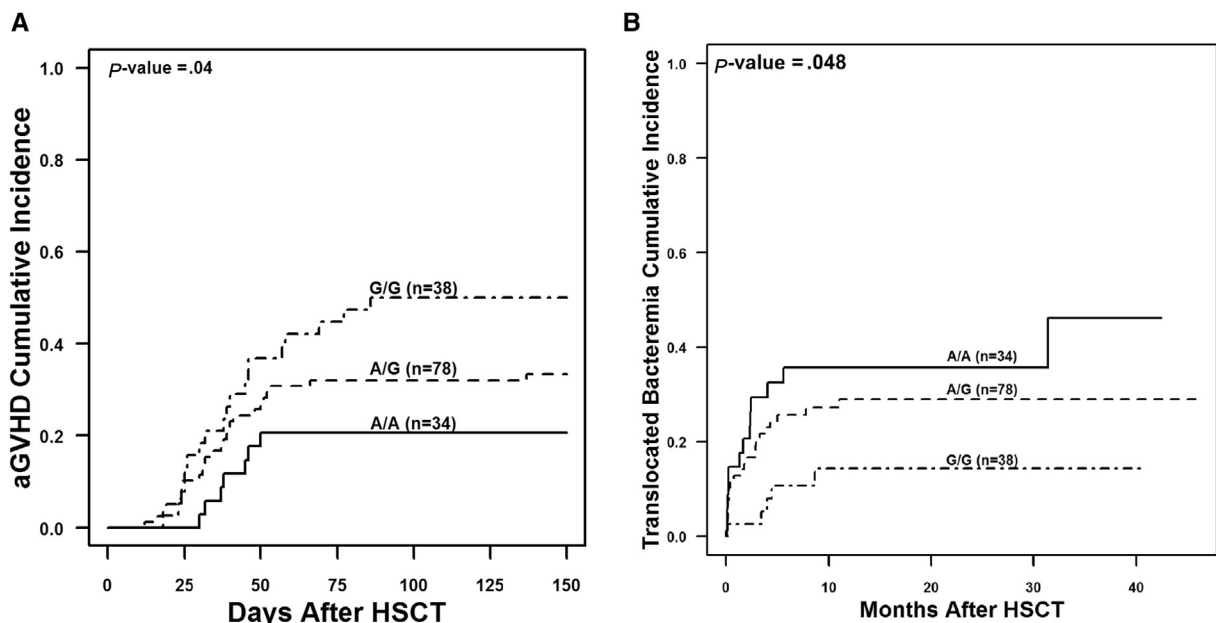


Figure 1. *FUT2* genotype and risk of acute GVHD and bacteremia after HSCT. (A) Cumulative incidence of acute GVHD by *FUT2* genotype. (B) Cumulative incidence of bacteremia by *FUT2* genotype.

Table 2
Multivariate Analysis of Risk Factors for GVHD

Variable	OR	P Value
Age	1.02	.20
<i>FUT2</i> genotype		
A/A	.42	.046
A/G	.71	.24
G/G	1.00	-
Myeloablative conditioning	2.00	.029
HLA match		
Matched sibling donor	.41	.026
Matched family or unrelated donor	.99	.98
Mismatched donor	1.00	-

Survival

One year after transplantation, 78.5% (95% confidence interval, 71.7% to 85.9%) of children were surviving and 69.6% (95% confidence interval, 60% to 80.7%) were surviving at 3 years. *FUT2* genotype did not affect overall survival (Figure 2).

DISCUSSION

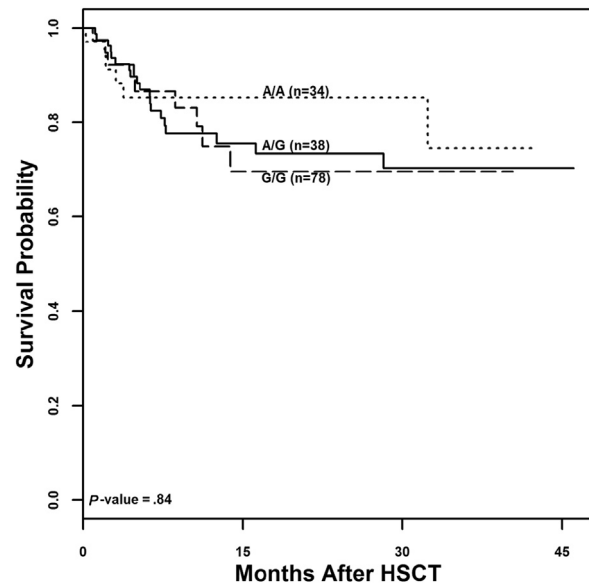
This is the first report describing the affect of *FUT2* genotype on outcome of allogeneic HSCT, to our knowledge. Our data show that the nonsecretor AA genotype is associated with a reduced risk of bacteremia, as originally hypothesized. In contrast, acute GVHD risk was decreased in children with the nonsecretor AA genotype, a result that is the opposite of our initial hypothesis.

The increased risk of infection associated with the nonsecretor genotype is a novel finding in the HSCT setting. However, our data are similar to a report by Morrow et al. in premature infants, showing that low or nonsecretor status strongly predicted risk of adverse outcomes that occurred after the first postnatal week, including overall mortality and death from sepsis and necrotizing enterocolitis [25]. Nonsecretor individuals may have reduced integrity of the mucosal barrier, allowing access of pathogens to the blood stream, as suggested by a recent report by Pham et al. showing that IL-22RA1-mediated fucosylation of gut epithelium via *FUT2* modifies epithelial translocation of *Enterococcus faecalis* [26]. The report of Pham et al. is in agreement with Maroni et al., who described a lack of fucosylated glycoproteins in the mucosal epithelium of nonsecretor individuals and associated modification of the microbiome [27].

The frequency of Crohn's disease is increased in nonsecretor individuals, leading us to expect that the risk of GVHD might be similarly increased. The increased risk of Crohn's disease appears to be mediated by changes in the microbiome associated with the nonsecretor genotype [10]. The human gut microbiome is largely made up of members of the

Table 3
Multivariate Analysis of Risk Factors for Bacteremia

Variable	OR	P Value
Age	1.03	.13
<i>FUT2</i> genotype		
A/A	4.01	.007
A/G	2.49	.066
G/G	1.00	-
Myeloablative conditioning	.82	.64
HLA match		
Matched sibling donor	.98	.96
Matched family or unrelated donor	.93	.85
Mismatched donor	1.00	-

**Figure 2.** Probability of overall survival by *FUT2* genotype.

phylae firmicutes and bacteroidetes. A number of authors have described changes in the microbiome associated with the nonsecretor genotype, in general showing reduced bacterial richness and depletion of firmicutes in nonsecretor individuals. Wacklin et al. (2011) studied normal volunteers and showed decreased diversity, richness, and abundance of bifidobacteria in nonsecretors [9]. In a subsequent study, Wacklin et al. (2014) showed lower microbial richness and increased representation of *Blautia*, *Clostridium sphenoides*, *Ruminococcus gnavus*, and *Dore formicigenerans* in nonsecretor individuals [28]. In agreement with this, Tong et al. studied 39 healthy subjects and showed depletion of firmicutes and expansion of proteobacteria in nonsecretors [11]. In a study that included persons with Crohn's disease and normal controls, Rausch et al. identified reduced prominence of *Lactobacillae* and *Stenotrophomonas* in nonsecretors [10]. In addition, *Lachnospiraceae*, *Prevotella*, *Brevundimonas*, *Sutterella*, and *Faecalibacterium* were present in the microbiome of nonsecretors while not present in secretors, representing organisms that may digest mucin and contribute to mucosal impairment. Jenq et al. have described changes in the gut microbiome (loss of diversity, depletion of clostridia, and expansion of lactobacillales—a member of the phylum firmicutes) in mouse models of acute GVHD and in human studies [1]. Taur et al. recently found that overall survival was significantly different among 3 groups defined by intestinal bacterial diversity after HSCT, whereas subjects with high diversity were more likely to have better overall survival [29]. The changes in the microbiome reported in these studies differ from those reported in association with the nonsecretor genotype. It is possible that decreased risk of GVHD in nonsecretor recipients is due to a change in the representation of specific groups of bacteria. Tong et al. note both compositional and functional changes in the microbiota of nonsecretors [11]. Some bacteroidetes species produce short-chain fatty acids, which are known to have anti-inflammatory effects, so it is possible that the nonsecretors have more beneficial anti-inflammatory commensals, explaining the unexpected finding of relative protection against GVHD. *FUT2* secretor status has been associated with increased susceptibility to norovirus and rotavirus [30–32]. It

remains possible that differential exposure to other pathogens in the gastrointestinal tract modifies the incidence of GVHD and contributes to the increased risk in secretor individuals. Alternatively, differential protein glycosylation might have a direct effect on alloreactivity. Recent xenotransplantation studies have suggested that variable glycosylation in the host modifies graft rejection in solid-organ transplantation mouse models [33].

The data reported in this study indicate an important role for FUT2 genotype in transplantation outcomes. However, we recognize that the FUT2 genotype may be only 1 of many factors (eg, diet, antibiotic exposure, conditioning regimen, etc.) that modify the gut microbiota in HSCT patients. Our study was focused on pediatric patients. The human gut microbiome reaches an adult state at about 2 years of age, so our data may be generalizable to adults, but our findings need to be tested in an adult study to determine this. Additional studies that directly address the composition of the microbiome in transplant recipients of each genotype are needed to explore the mechanism of the observed effects.

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Authorship contribution: A.R. surveyed the literature, interpreted the data, and wrote the paper; A.L.M. assisted with study design and data interpretation, L.R.P. and K.E.L. performed the genotyping, A.L. performed statistical analysis and generated the figures, and S.M.D. designed and directed the research, assisted with data interpretation, and reviewed/edited the manuscript.

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